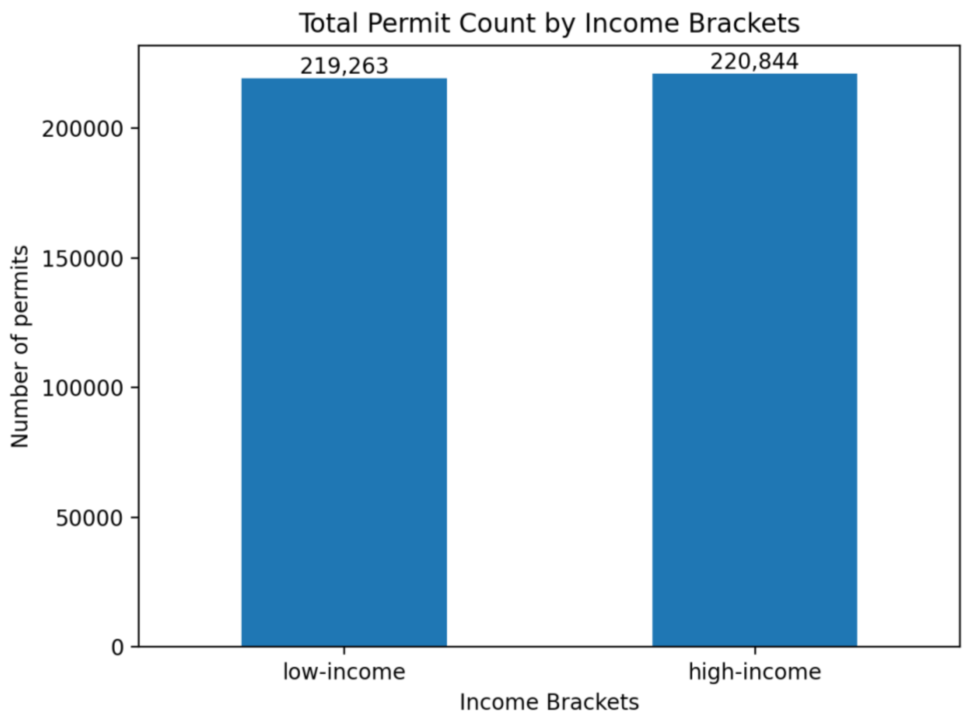
Income Inequality in Los Angeles Building Permit Activity

Hairong(Rona) Zhang

How does permit activity differ between high-income and low-income census tracts? Are certain improvement types (e.g., energy efficiency upgrades, accessibility modifications) more or less common in different income brackets?

Graph 1



Python as cell1

```
# Core Python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import streamlit as st

from snowflake.snowpark.context import get_active_session
session = get_active_session()
```

Python as cell2

```
session.sql("USE ROLE TRAINING_ROLE").collect()
session.sql("USE DATABASE LA_PERMIT_DATA").collect()
session.sql("USE SCHEMA PUBLIC").collect()
```

	status
0	Statement executed successfully.

Python as cell3

```
records = session.table("PERMIT_RECORDS").to_pandas()
records.head()
```

	ASSESSOR_BOOK	ASSESSOR_PAGE	ASSESSOR_PARCEL	TRACT	BLOCK	LOT	REFERENCE_NUM	PC
0	4211	033	034	TR 60110-01_REC-C	None	2	13LA	130
1	2533	029	032	TR 29370	None	4	13WL55244	130
2	5486	025	031	LEWIS TRACT	B	30	13LA25080	130
3	5537	009	018	ZAHN TRACT	None	14	13LA25105	130
4	2217	030	030	TR 1000	None	222	13LA25134	130

Python as cell11 0.3s

```
census_tracts = session.table("CENSUS_TRACTS").to_pandas()
census_tracts.head()
```

	CENSUS_TRACT	MED_HH_INCOME	NUM_HOUSEHOLDS	AMI_CATEGORY	BELOW_AMI	BELOW_60PCT_AMI	BELO
0	6,037,101,110	84,091	1,558	Low Income	☑	☐	
1	6,037,101,122	99,583	1,407	Low Income	☐	☐	
2	6,037,101,220	69,676	1,357	Low Income	☑	☑	
3	6,037,101,221	53,798	1,483	Very Low Income	☑	☑	
4	6,037,101,222	45,662	948	Very Low Income	☑	☑	

+ Python + SQL + Markdown

Python as cell6

```python
# Create High and Low Income Groups
records_with_census["MED_HH_INCOME"] = pd.to_numeric(
    records_with_census["MED_HH_INCOME"],
    errors="coerce"
)

# Drop rows without income
records_with_census = records_with_census.dropna(
    subset=["MED_HH_INCOME"]
)

# Median split: bottom 50% = Low, top 50% = High
median_cutoff = records_with_census["MED_HH_INCOME"].median()

records_with_census["INCOME_BRACKETS"] = np.where(
    records_with_census["MED_HH_INCOME"] >= median_cutoff,
    "high-income",
    "low-income"
)

records_with_census["INCOME_BRACKETS"].value_counts()
```

INCOME_BRACKETS	count
high-income	221,126
low-income	219,488

Python as cell4

```python
# Convert date columns
for col in ["ISSUE_DATE", "STATUS_DATE"]:
    records[col] = pd.to_datetime(records[col], errors="coerce")

# Clean valuation (string → numeric)
records["VALUATION"] = pd.to_numeric(
        records["VALUATION"].str.replace("$", "").str.replace(",",""),
        errors="coerce"
    )

# Create CT key for joining to census
ct_numeric = pd.to_numeric(records["CENSUS_TRACT"], errors="coerce")
records["CT"] = (ct_numeric * 100) + 6037000000
```

Python as cell5 1.9s

```python
# Join Permits and Census Tracts
records_with_census = records.merge(census_tracts, left_on="CT", right_on="CENSUS_TRACT", how="inner
records_with_census.head()
```

	ASSESSOR_BOOK	ASSESSOR_PAGE	ASSESSOR_PARCEL	TRACT	BLOCK	LOT	REFERENCE_NUM	PCIS_PE
0	2533	029	032	TR 29370	None	4	13WL55244	13016-3
1	5486	025	031	LEWIS TRACT	B	30	13LA25080	13014-1
2	5537	009	018	ZAHN TRACT	None	14	13LA25105	13020-1
3	2217	030	030	TR 1000	None	222	13LA25134	13019-1
4	2167	001	015	P M 1055	None	B	None	13041-1

Python as cell7

0.0s

```python
# Classify Improvement Types from AI_DESCRIPTION
def improvement_types(text):
    text = text.lower()

    # Energy efficiency
    if "solar" in text or "hvac" in text or "heat pump" in text:
        return "Energy efficiency upgrades"

    # Accessibility
    elif "ada" in text or "ramp" in text or "wheelchair" in text:
        return "Accessibility modifications"

    # Repairs / Maintenance
    elif "roof" in text or "repair" in text or "replace" in text:
        return "Repair / Maintenance"

    # Interior remodels
    elif "kitchen" in text or "bath" in text:
        return "Interior remodel"

    # ADU
    elif "adu" in text or "accessory dwelling" in text:
        return "ADU"

    else:
        return "Other"
```

Python as cell8

```python
records_with_census["IMPROVEMENT_TYPE"] = (
    records_with_census["AI_DESCRIPTION"].fillna("")
    .apply(improvement_types)
)

records_with_census["IMPROVEMENT_TYPE"].value_counts()
```

IMPROVEMENT_TYPE	count
Other	337,235
Repair / Maintenance	41,403
ADU	21,668
Interior remodel	20,356
Energy efficiency upgrades	18,351
Accessibility modifications	1,601

Python as cell9

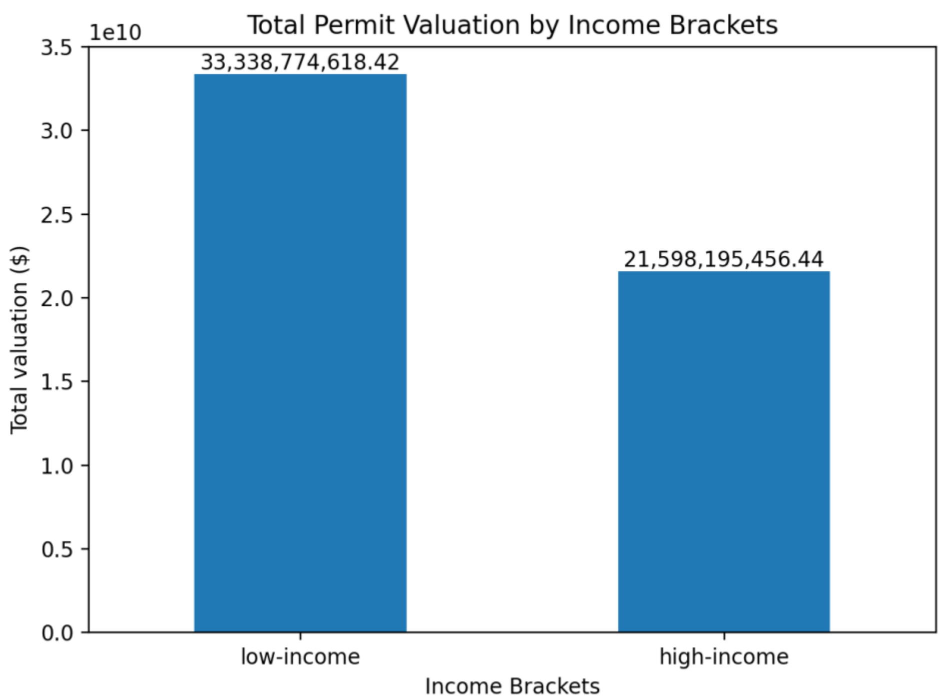
```python
records_with_census = records_with_census.dropna(
    subset=["ISSUE_DATE", "INCOME_BRACKETS"]
)
```

Python as cell10 0.4s

```python
# Graph 1. Total Permit Count by Income Brackets
permit_counts = (
    records_with_census
        .groupby("INCOME_BRACKETS")["PCIS_PERMIT_NUM"]
        .nunique()
        .reindex(["low-income", "high-income"])
)

plt.figure()
ax = permit_counts.plot(kind="bar")
plt.title("Total Permit Count by Income Brackets")
plt.xlabel("Income Brackets")
plt.ylabel("Number of permits")
plt.xticks(rotation=0)
for i, value in enumerate(permit_counts):
    ax.text(i, value, f"{value:,}", ha="center", va="bottom")
plt.tight_layout()
plt.show()
```

This graph shows the total number of building permits issued in low-income versus high-income census tracts in Los Angeles. The permit counts are very similar across the two groups, indicating that overall construction activity is occurring at comparable volumes in both low-income and high-income areas. However, this similarity in volume does not necessarily imply equality in investment or scale. This graph establishes an important baseline for the overall story that while permit activity appears evenly distributed in quantity, later graphs reveal that the types of improvements and the economic value of those permits differ substantially by income level, highlighting deeper structural inequality.

Graph 2

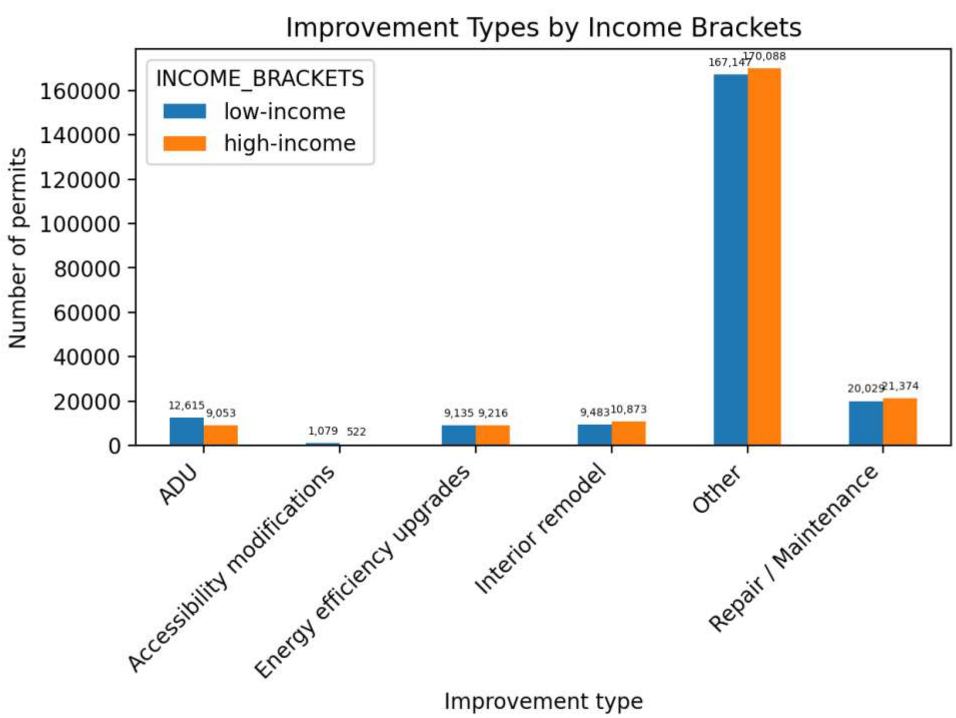


Python as cell12

```python
# Graph 2. Total Permit Valuation by Income Brackets
valuation_by_income = (
    records_with_census
        .groupby("INCOME_BRACKETS")["VALUATION"]
        .sum()
        .reindex(["low-income", "high-income"])
)

plt.figure()
ax=valuation_by_income.plot(kind="bar")
plt.title("Total Permit Valuation by Income Brackets")
plt.xlabel("Income Brackets")
plt.ylabel("Total valuation ($)")
plt.xticks(rotation=0)
for i, value in enumerate(valuation_by_income):
    ax.text(i, value, f"{value:,}", ha="center", va="bottom")
plt.tight_layout()
plt.show()
```

This graph shows the total dollar value of building permits issued in low-income versus high-income census tracts. Unlike the similar permit counts seen in Graph 1, this chart reveals a large valuation gap, with low-income areas accounting for significantly higher total permit valuation than high-income areas. This result suggests that while permit volume is similar, investment patterns differ in scale and purpose, likely driven by large developments or major infrastructure projects in lower-income neighborhoods. This graph deepens the story by showing that economic impact is not evenly distributed even when permit counts appear equal.

Graph 3

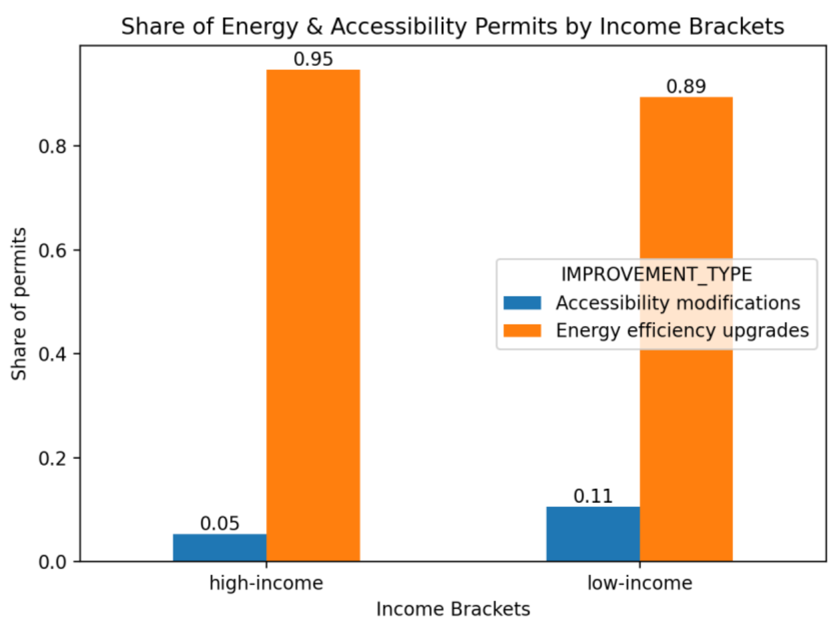


Python as cell13

```python
# Graph 3. Improvement Types by Income Brackets
improvement_counts = (
    records_with_census
        .groupby(["IMPROVEMENT_TYPE", "INCOME_BRACKETS"])["PCIS_PERMIT_NUM"]
        .count()
        .unstack(fill_value=0)
)

improvement_counts = improvement_counts[["low-income", "high-income"]]

plt.figure()
ax=improvement_counts.plot(kind="bar")
plt.title("Improvement Types by Income Brackets")
plt.xlabel("Improvement type")
plt.ylabel("Number of permits")
plt.xticks(rotation=45, ha="right")
for container in ax.containers:
    labels = [f"{int(bar.get_height()):,}" for bar in container]
    ax.bar_label(
        container,
        labels=labels,
        fontsize=5,
        padding=3
    )

plt.tight_layout()
plt.show()
```

This graph compares the distribution of different improvement types across income groups. The results show that the “Other” category dominates permit activity for both low-income and high-income areas, indicating that most permits fall outside the main classified improvement types. While both groups show similar patterns across categories, high-income areas consistently issue slightly more permits across most improvement types, including interior remodels and repairs. This suggests that although development types are broadly similar, high-income neighborhoods maintain a modest but consistent advantage in renovation activity.

Graph 4

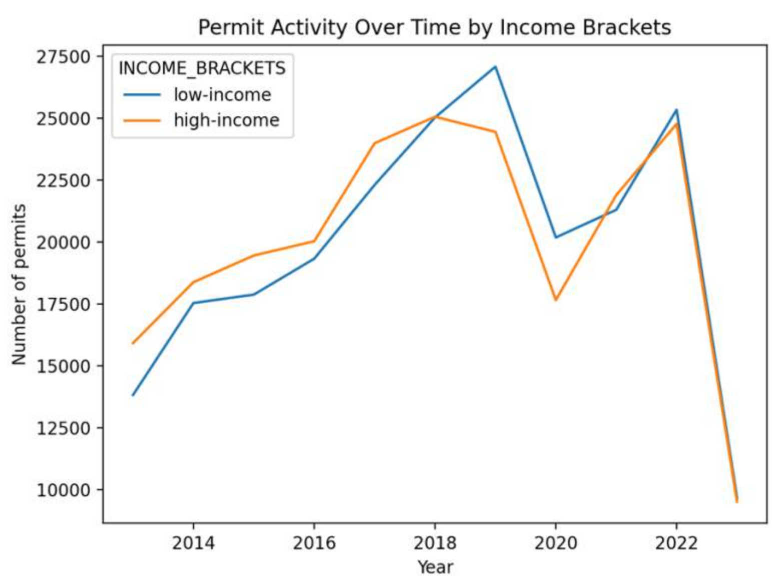


Python as cell14 0.5s

```python
# Graph 4. Share of Energy & Accessibility Permits by Income Brackets

# Filter only Energy & Accessibility permits
subgroup = records_with_census[
    records_with_census["IMPROVEMENT_TYPE"].isin([
        "Energy efficiency upgrades",
        "Accessibility modifications"
    ])
]

subgroup_counts = (
    subgroup
    .groupby(["INCOME_BRACKETS", "IMPROVEMENT_TYPE"])["PCIS_PERMIT_NUM"]
    .count()
    .unstack(fill_value=0)
)

# Column-by-column division
row_totals = subgroup_counts.sum(axis=1)
share = subgroup_counts.copy()
for col in share.columns:
    share[col] = share[col] / row_totals

plt.figure()
ax=share.plot(kind="bar")
plt.title("Share of Energy & Accessibility Permits by Income Brackets")
plt.xlabel("Income Brackets")
plt.ylabel("Share of permits")
plt.xticks(rotation=0)
for container in ax.containers:
    ax.bar_label(container, fmt="%.2f", label_type="edge")
plt.tight_layout()
plt.show()
```

This graph focuses specifically on energy efficiency upgrades and accessibility modifications, displaying each income group’s share of these socially impactful improvements. The results show that high-income areas devote a larger proportion of their improvement activity to energy efficiency upgrades, while low-income areas allocate a greater share toward accessibility modifications. This pattern suggests that higher-income communities are more able to invest in long-term sustainability improvements, whereas lower-income communities prioritize immediate functional and mobility-related needs. So this highlights how income level shapes the type of improvements households prioritizes.

Graph 5



Python as cell15 0.6s

```python
# Graph 5. Permit Trends Over Time by Income Brackets

# Extract year
records_with_census["YEAR"] = records_with_census["ISSUE_DATE"].dt.year

yearly_trend = (
    records_with_census
        .groupby(["YEAR", "INCOME_BRACKETS"])["PCIS_PERMIT_NUM"]
        .count()
        .unstack(fill_value=0)[["low-income", "high-income"]]
)

plt.figure()
yearly_trend.plot()
plt.title("Permit Activity Over Time by Income Brackets")
plt.xlabel("Year")
plt.ylabel("Number of permits")
plt.tight_layout()
plt.show()
```

This graph shows how permit activity has changed over the years for low-income and high-income census tracts. While both groups follow similar overall growth trends, differences in volatility and growth pace indicate that development momentum is not perfectly synchronized across income levels. Periods of rapid growth tend to favor one group more than the other, suggesting that economic cycles and policy shifts do not impact all communities equally. This graph completes the story by showing that inequality is not only spatial but also temporal, evolving over time rather than remaining static.

Dashboard

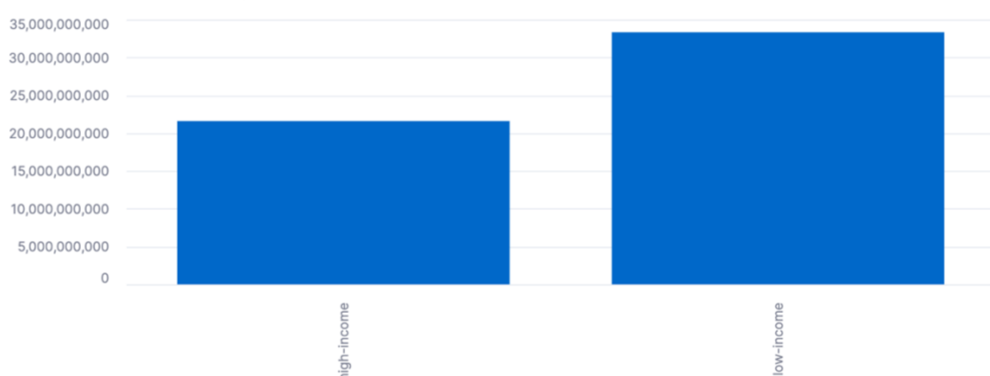
LA Building Permit Inequality Dashboard

Decision-focused view of permit activity across income brackets (low vs high)

Key Metrics

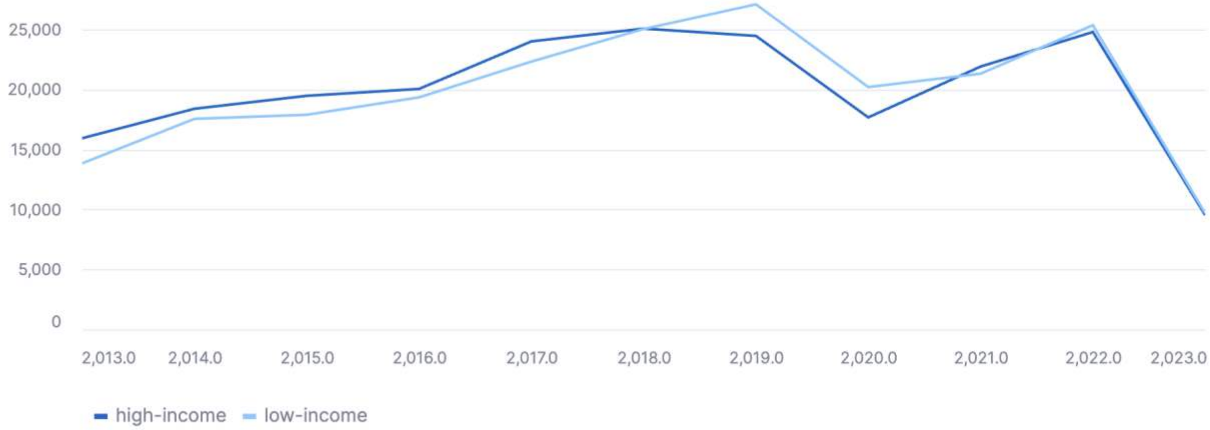
Total Permits	Low-Income Permits	High-Income Permits
440,614	219,488	221,126

Total Permit Valuation by Income Brackets



This chart shows total dollar valuation of permitted construction in low and high income census tracts.

Permit Trends Over Time by Income Brackets



This chart shows how permit activity has evolved over time by income brackets.

Python as cell16 2.4s

```python
# Dashboard
st.title("LA Building Permit Inequality Dashboard")
st.write("Decision-focused view of permit activity across income brackets (low vs high)")

base_df = records_with_census.copy()

# Sidebar Filters
st.sidebar.header("Filters")

income_filter = st.sidebar.multiselect(
    "Select Income Brackets:",
    options=sorted(base_df["INCOME_BRACKETS"].unique()),
    default=sorted(base_df["INCOME_BRACKETS"].unique())
)

df = base_df[base_df["INCOME_BRACKETS"].isin(income_filter)]

# Year column
df["YEAR"] = df["ISSUE_DATE"].dt.year

year_filter = st.sidebar.slider(
    "Select Year Range:",
    int(base_df["ISSUE_DATE"].dt.year.min()),
    int(base_df["ISSUE_DATE"].dt.year.max()),
    (
        int(base_df["ISSUE_DATE"].dt.year.min()),
        int(base_df["ISSUE_DATE"].dt.year.max())
    )
)

df = df[(df["YEAR"] >= year_filter[0]) & (df["YEAR"] <= year_filter[1])]

st.sidebar.write(f"Records: {len(df):,}")
```

```python
st.subheader("Key Metrics")

total_permits = df["PCIS_PERMIT_NUM"].count()
low_count = df[df["INCOME_BRACKETS"] == "low-income"]["PCIS_PERMIT_NUM"].count()
high_count = df[df["INCOME_BRACKETS"] == "high-income"]["PCIS_PERMIT_NUM"].count()

col1, col2, col3 = st.columns(3)
col1.metric("Total Permits", f"{total_permits:,}")
col2.metric("Low-Income Permits", f"{low_count:,}")
col3.metric("High-Income Permits", f"{high_count:,}")

# GRAPH 1 - TOTAL PERMIT VALUATION (UPDATED)

st.subheader("Total Permit Valuation by Income Brackets")

g1_valuation = (
    df
    .groupby("INCOME_BRACKETS")["VALUATION"]
    .sum()
)

st.bar_chart(g1_valuation)

st.caption(
    "This chart shows total dollar valuation of permitted construction "
    "in low and high income census tracts."
)

# GRAPH 2 - PERMIT TRENDS OVER TIME
st.subheader("Permit Trends Over Time by Income Brackets")

g2 = (
    df
    .groupby(["YEAR", "INCOME_BRACKETS"])["PCIS_PERMIT_NUM"]
    .count()
    .unstack(fill_value=0)
)

st.line_chart(g2)

st.caption(
    "This chart shows how permit activity has evolved over time by income brackets."
)
```

Overall, across Los Angeles census tracts, permit activity between high-income and low-income areas is similar in volume, indicating that construction and renovation occur at comparable frequencies across income groups. However, important differences emerge in investment scale and the types of improvements pursued. While total permit counts are balanced, low-income census tracts show a higher total permit valuation than high-income areas, indicating that more expensive or large-scale projects may be concentrated in lower-income communities. Differences in improvement composition further illustrate this divide. High-income communities devote a greater share of permits to energy efficiency upgrades, reflecting stronger capacity for long-term sustainability investments, whereas low-income communities show a higher proportion of accessibility modifications, suggesting a focus on immediate functional and mobility needs. Over time, both groups follow similar growth trends in permitting, but these structural differences in valuation and improvement priorities persist. So in conclusion, the findings demonstrate that income level shapes not how much building occurs, but how resources are allocated and what kinds of improvements are prioritized.